BUS 9430 · CAPSTONE 2 · DATA UNDERSTANDING AND PREPARATION

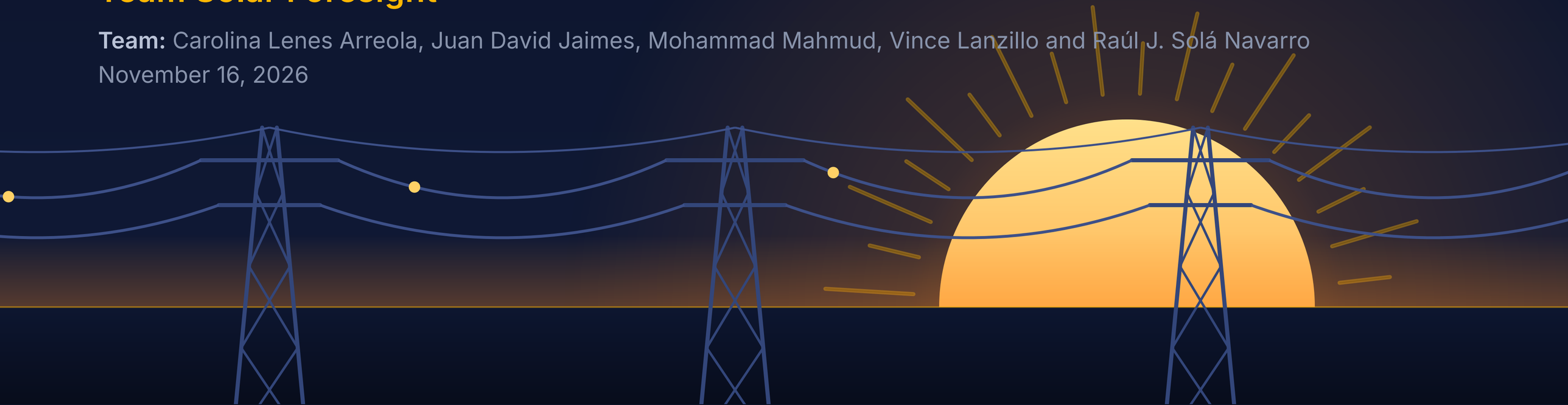
Tomorrow's Sun, Today's Plan

What our data shows, and how we get it ready

Team Solar Foresight

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November 16, 2026



THE BUSINESS PROBLEM

Solar is the part of tomorrow's plan that depends on the weather

WHAT CAISO DOES

Locks in tomorrow's power plan today

Bids close at 10 a.m. Pacific. The plan is published at about 1 p.m. It says which plants run tomorrow and how much each makes.

WHAT GOES WRONG

The solar forecast misses

Too high, and operators buy backup power at the last minute. Too low, and they hold reserves they never use.

WHAT WE DO

Forecast solar output a day ahead

Hour by hour, using only what is known at 10 a.m. We then compare ourselves with CAISO's own forecast.

WHAT WE FORECAST

We forecast solar output, not electricity demand

TOTAL DEMAND
What customers use

−

WIND AND SOLAR
What the weather supplies

=

NET LOAD
What gas, batteries and imports must cover

DEMAND

CAISO already forecasts it

It is a different problem, and the operator has its own tools for it.

OUR TARGET VARIABLE

Hourly solar output, in megawatts (MW)

How much power utility-scale solar plants in the CAISO area actually produce each hour, as measured by the U.S. Energy Information Administration.

Net load definition: U.S. Energy Information Administration (EIA).

THE DATA

Five public sources, one hourly table

ROLE	SOURCE	WHAT IT GIVES US	WHAT WE PULLED
Answer key	EIA Hourly Electric Grid Monitor	Actual hourly solar output for the CAISO area	Hourly, Jan 2024 to Oct 17, 2026
Inputs	Open-Meteo Previous Runs	Weather forecasts published before the 10 a.m. bid close	51 locations across California, Mar 2024 to Oct 2026
Benchmark	CAISO OASIS, report SLD_REN_FCST	CAISO's own day-ahead solar forecast	Hourly, same dates, three solar regions added up
Scale	EIA Form 860M	Installed solar capacity, so errors compare fairly over time	Monthly, Mar 2024 to Sep 2026
Flags	CAISO curtailment files	Solar that was turned down because the grid could not use it	Mar 2024 to Oct 2026

Weather data by Open-Meteo.com (CC BY 4.0). All data files are saved in the team repository.

THE RULE THAT KEEPS THE TEST HONEST

We may only use what is known at 10 a.m.

ALLOWED (MAIN MODEL)

Weather forecast made two days before

Published before the 10 a.m. bid close for every hour of the next day. This is the input we use.

COMPARISON ONLY

Weather forecast made one day before

For afternoon hours it is likely published after the bid close. We run it only as a labeled comparison, never as our result.

Why it matters: a forecast that peeks at later information looks better than it could be in real life.

Both versions are in the data, so the comparison is a single switch.

DATA UNDERSTANDING

The data at a glance: nearly complete

PERIOD COVERED

Mar 2024 to Oct 2026

954 days, from March 8, 2024 to October 17, 2026.

HOURS OF DATA

22,896

One row per hour. 11,733 of them are daylight hours, when solar can make power.

HOURS WITH A GAP

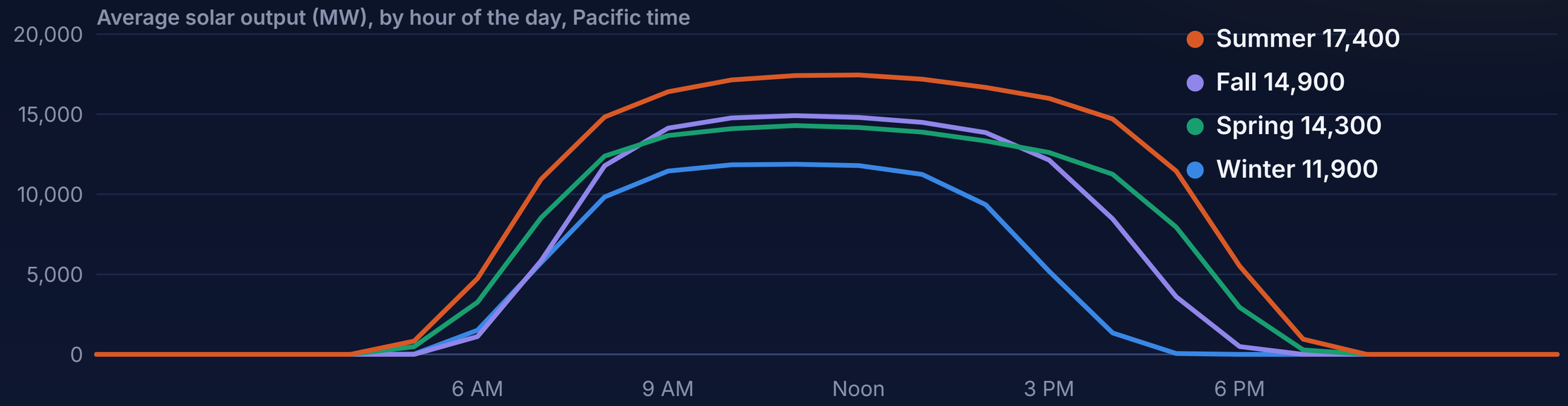
< 1%

225 hours across all sources, out of 22,896. Gaps are in the weather feed (129), the CAISO forecast (72) and EIA (24).

So the data is in good shape before we change anything.

DATA UNDERSTANDING

A typical solar day: up at 6, peak at noon, done by 8



WHAT WE SEE

Summer peaks about 47% higher than winter

About 17,400 MW at midday in summer, versus 11,900 MW in winter.

WHAT IT MEANS FOR THE MODEL

Hour of day and season matter

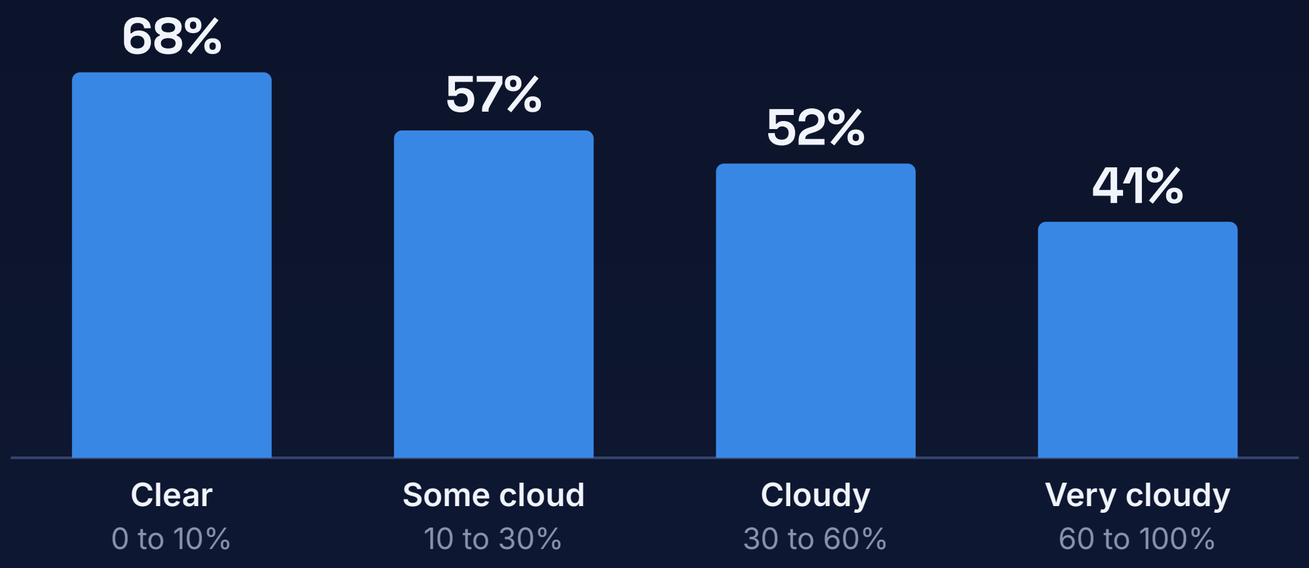
The model has to know both, so we give them to it directly.

Average over March 2024 to October 2026. Winter is December to February. Source: EIA.

DATA UNDERSTANDING

Cloudier forecast, less solar

Midday output, % of installed solar capacity, by forecast cloud cover



0.87

How closely forecast sunshine tracks actual output, on a scale where 1.0 is perfect. Measured across 9,191 daylight hours.

54%

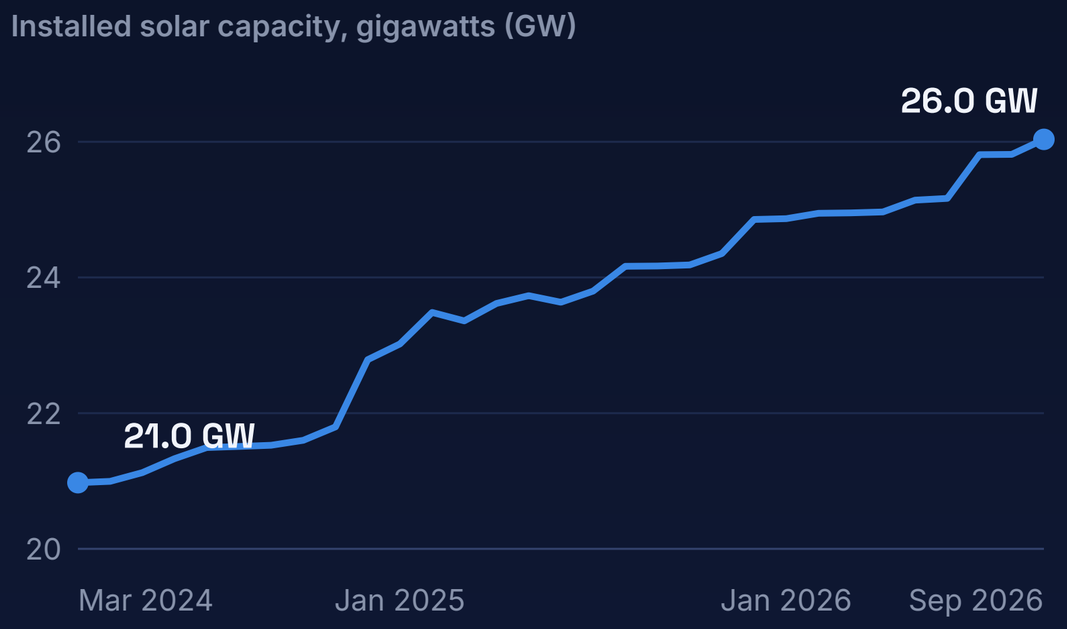
Biggest one-day drop: on October 10, 2026, output fell from 161 to 73 gigawatt-hours (GWh) in a day.

Midday = 10 a.m. to 1 p.m. Pacific. Cloud cover is the forecast made two days earlier. Includes all seasons, so winter's lower sun also plays a part.
Sources: EIA, Open-Meteo.

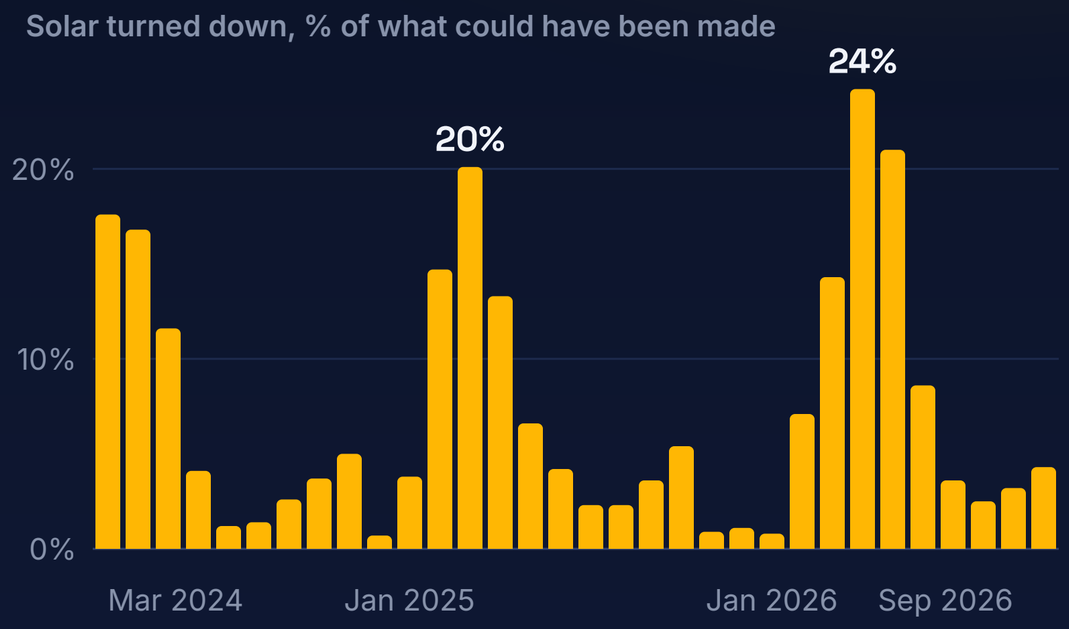
DATA UNDERSTANDING

More panels every month, and more solar turned down

Installed solar capacity grew 24%



Spring is when solar gets turned down



WHY IT MATTERS

Errors must be scaled by capacity

A 1,000 MW miss meant more in 2024 than it will in 2026. We report errors as a share of installed capacity.

WHY IT MATTERS

Turned-down solar hides true output

In April 2026 about 1 in 4 possible megawatt-hours was turned down. Over the whole period it is 7.6%.

Sources: EIA Form 860M (capacity, includes about 2.8 GW in Arizona and Nevada); CAISO curtailment reports.

DATA UNDERSTANDING

Four data problems, and what we do about each

ISSUE	WHAT WE FOUND	WHAT WE DO
Clocks disagree	EIA reports in UTC. CAISO files use the local clock, which shifts with daylight saving time.	Convert every source to one clock (hour ending, UTC) and check against sunrise and sunset.
Missing hours	225 of 22,896 hours (under 1%): weather 129, CAISO forecast 72, EIA 24.	Fill short gaps, drop long ones, and say which.
Turned-down solar	7.6% of possible solar energy over the period. Up to 24% in a spring month.	Flag those hours and report results with and without them.
More panels over time	Installed capacity up 24%, from 21.0 to 26.0 GW.	Scale errors by installed capacity so months compare fairly.

Also noted: tiny negative readings at night (about 40 MW below zero, against peaks of 17,000 MW). They are handled explicitly.

PRIVACY AND DELIVERY RISKS

Low privacy risk, three delivery risks

PRIVACY: LOW

Public, system-wide data

No customers, households or personal information. The capacity file names power plants and companies, which are public records.

Open-Meteo requires credit (CC BY 4.0) and its free tier is non-commercial. EIA and CAISO data are public, and we cite them.

DELIVERY RISK 1

Timing at 10 a.m.

A forecast is only useful if its inputs exist by the bid close. A late weather feed would break it.

We use the two-day-ahead forecast and check the one-day version only as a comparison.

DELIVERY RISK 2 AND 3

Limits and reproducibility

The free weather service caps daily calls. Our full pull used about 6,500 of the 10,000 allowed in a day.

Raw files are saved in the repository with a data dictionary, so nobody needs to re-pull.

THE BAR TO BEAT

How good is CAISO’s own forecast on our data?

SAME AS YESTERDAY

4.9%

Average miss as a share of installed capacity. The floor to beat. It is 6.1% if limited to what is known at 10 a.m.

CAISO, AS PUBLISHED

5.7%

Runs about 830 MW too high on average, because it plans for solar that is then turned down.

CAISO, ADJUSTED

3.7%

With turned-down solar added back. The fairest picture of CAISO's skill. The gap to actual output almost disappears.

OUR TARGET: NO MORE THAN 1.25 TIMES CAISO'S MISS

That means 4.6% to about 7%, depending on which CAISO number counts

We will fix this before modeling and show both numbers.

Scored on 9,133 daylight hours over 758 days in the training and validation days (March 2024 to April 9, 2026). The test days stay locked.

DATA PREPARATION

From five sources to one table

1

Pull

Download each source for the same dates.

2

Align

One clock, one row per hour.

3

Clean

Handle gaps and flag curtailed hours.

4

Add features

Hour, season, sun angle, scaled by capacity.

5

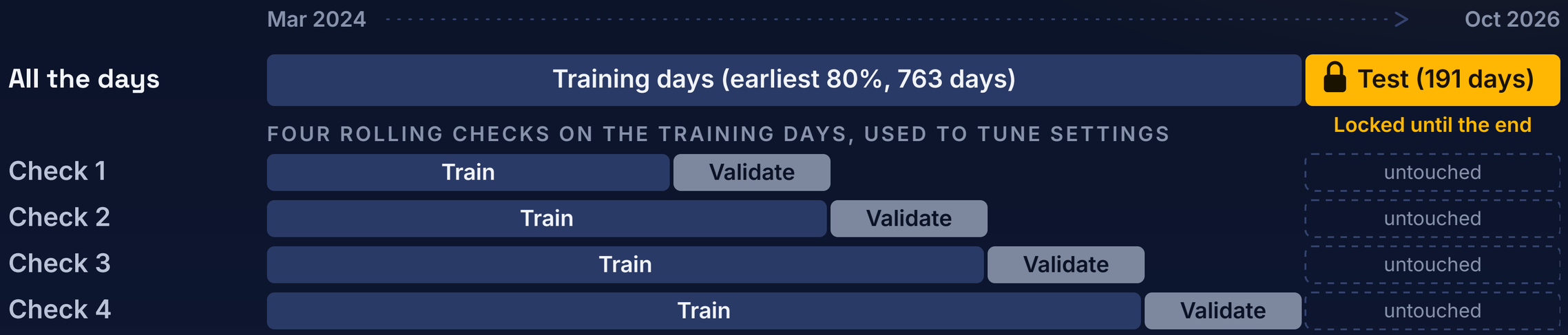
Split

Time-ordered training, validation and test sets.

Steps 1 and 2 are done: 22,896 aligned hourly rows. Steps 3 to 5 follow the rules on the previous slides.

DATA PREPARATION

Test on days the model has never seen



1

Train

The earliest 80% of days (763 days, to April 9, 2026) teach the models.

2

Validate

Four rolling checks of about three months each compare the models and tune them.

3

Test

The latest 191 days (April 10 to October 17, 2026), scored once at the end. No winter, so winter is judged in validation.

WHAT COMES NEXT

We choose models after we see the data

STEP 1
Score the two reference forecasts

→

STEP 2
Explore the data

→

STEP 3
Shortlist candidate models

→

STEP 4
Compare on the same folds

JUDGED ON
Average miss compared with CAISO

JUDGED ON
Worst-case overshoot, which drives backup power

JUDGED ON
How easily we can explain it

Step 1 is done: see the bar to beat.

RISKS AND OPEN DECISIONS

Four things we are watching

SHORT HISTORY, NO WINTER IN THE TEST

About two and three-quarter years

Each calendar month appears two to three times. The test block runs April to October, so we judge winter in validation and say so plainly.

WHICH YARDSTICK

CAISO's number depends on turned-down solar

Its miss is 5.7% as published and 3.7% with turned-down solar added back. We report both and fix the target before modeling.

WEATHER TIMING

Which forecast is available at 10 a.m.

Main model uses the forecast made two days before. The one-day version is a labeled comparison.

TURNED-DOWN SOLAR

Concentrated in spring, inside our test block

April and May 2026 saw 21% to 24% turned down. We flag those hours and report results with and without them.

NEXT STEPS

From a clean table to results

Nov 17–27

Explore the data and compare candidate models.

Nov 30–Dec 8

Score the models against CAISO and build the charts.

Dec 14

Capstone 3: models and results.

Your input: Is there anything in the data or the test design you would want us to check before we build models?

APPENDIX A

Data dictionary: the main columns

COLUMN	MEANING	UNIT	SOURCE
`hour_end_utc`	The hour ending at this time, UTC (the key for joining all sources)	timestamp	All, aligned
`eia_mw`	Actual utility-scale solar output (target variable)	MW	EIA
`caiso_dam_mw`	CAISO day-ahead solar forecast (benchmark only)	MW	CAISO OASIS
`capacity_mw`	Installed solar capacity in that month	MW	EIA Form 860M
`cloud_cover`, `shortwave_radiation`, `temperature_2m`, …	Weather forecast inputs, two-day-ahead version, capacity-weighted across locations	%, W/m², °C	Open-Meteo
`curt_mwh`	Solar energy turned down in that hour	MWh	CAISO curtailment files

Full list of columns, units and checks: docs/DATA_DICTIONARY.md in the team repository.

APPENDIX B

Sources

U.S. EIA, Hourly Electric Grid Monitor and API; net load definition:
eia.gov/opendata

U.S. EIA, Form 860M, monthly installed capacity:
eia.gov/electricity/data/eia860m

Open-Meteo, Previous Runs API, CC BY 4.0:
open-meteo.com/en/docs/previous-runs-api

CAISO, day-ahead market (bids close 10 a.m. Pacific, results about 1 p.m.):
caiso.com/market-operations/products-services

CAISO OASIS, day-ahead solar forecast, report SLD_REN_FCST:
oasis.caiso.com

CAISO, production and curtailments data (curtailment workbooks and daily renewables report):
caiso.com/library/production-curtailments-data

Team repository, raw data, data dictionary, FAQ and how-to:
github.com/RaulSolaNavarro/caiso-solar-forecast